are peacefully “turned off” in the vacuum state. As far as quantum field theory is concerned, a human being or the center of a star isn’t all that different from empty space.

Maybe the fact that the entropy of a black hole is proportional to its area is just what we should expect. In quantum field theory it’s natural for regions of space to have an entropy proportional to their boundary area, and a black hole is just a region of space. But a problem lurks beneath the surface. It’s natural for a region of space *in the vacuum state* to have an entropy proportional to its boundary area. But a black hole isn’t part of the vacuum state; there’s a black hole there, and spacetime is noticeably curved.

Black holes have a very special property: they represent the *highest-entropy states* we can have in any given size region of space. This provocative fact was first noticed by Bekenstein, and later refined by Raphael Bousso. If you start from a region within the vacuum state and try to increase its entropy, you must also increase its energy. (Since you started in the vacuum, there’s nowhere for the energy to go but up.) As you keep throwing in entropy, the energy also increases. Eventually you have so much energy in a fixed region that the whole thing can’t help but collapse into a black hole. That’s the limit; you can’t fit any more entropy into a region than you would have if a black hole were there.

That conclusion is profoundly different from what we would expect in an ordinary quantum field theory without gravity. There, there is no limit on how much entropy we can fit in a region, because there’s also no limit on how much energy there can be. This reflects the fact that there are an infinite number of degrees of freedom in quantum field theory, even in a finite-sized region.

Gravity appears to be different. There is a maximum amount of energy and entropy that can fit into a given region, which seems to imply that there are only a finite number of degrees of freedom there. Somehow these degrees of freedom become entangled in the right way to stitch together into the geometry of spacetime. It’s not just black holes: every region of spacetime has a maximum entropy we could imagine fitting into it (the entropy that a black hole of that size would have), and therefore a finite